

The Missing Rung: AI and the Entry-Level Tech Pipeline

Jennifer Kumura

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The Challenge

Generative AI is changing the economics of entry-level work. Tasks that once gave junior designers and engineers their first opportunity to learn—producing design assets, writing boilerplate code, and producing first-draft interfaces—can increasingly be automated or completed with AI assistance. The result may be less visible in total employment than in the pipeline into experienced work: if firms can obtain junior-level output without hiring juniors, fewer workers enter the profession and fewer eventually become experienced workers.

The question is therefore not simply whether AI reduces demand for junior workers, but whether the market will correct the resulting training shortfall before missing junior cohorts become a shortage of experienced talent.

This paper examines four occupations with different expected AI exposure: Graphic Designers, Software Developers, Web & Digital Interface Designers, and Home Health Aides. Software Developers provide a benchmark (Brynjolfsson et al., 2026); Graphic Designers provide the primary comparison; Interface Designers provide a more augmentation-oriented contrast; and Home Health Aides provide a lower-exposure control. The analysis does not claim that AI caused all observed employment changes.

The Economics

Three economic concepts explain why the entry-level pipeline can weaken.

First, training creates a positive externality. A firm bears the cost of developing a junior worker but may not capture the full return if that worker later leaves for another employer. Firms therefore have an incentive to invest less in training than would be optimal for the industry as a whole.

Second, the problem reflects short-run versus long-run elasticity. Firms can reduce junior hiring quickly when technology changes, but the supply of experienced workers cannot

respond as quickly because experience takes time to develop. A firm can eliminate a junior position today but cannot create an experienced replacement tomorrow.

Third, the specificity rule suggests targeting the market failure rather than the technology. If the problem is underinvestment in training, the response should restore investment in early-career development rather than restrict AI adoption.

Analysis

The first test asks whether the observed early-career employment pattern follows the predicted AI-exposure mechanism.

Using IPUMS Current Population Survey data (Flood et al., 2022), I compared employment among 22–25-year-olds in Graphic Designers and Web & Digital Interface Designers between a pooled baseline of 2022 Q1–2023 Q4 and a current period of 2024 Q4–2026 Q3. Graphic Designers declined 38.3%, while Interface Designers declined 51.3%. The difference was –13.0 percentage points, with a 95% confidence interval of –38.6 to +12.6 points.

This does not support the original hypothesis that Graphic Designers would experience the larger decline: the confidence interval includes zero, the point estimate runs opposite the prediction, and the small Interface Designer sample makes the estimate imprecise. The result is therefore inconclusive rather than evidence against AI disruption generally.

The Home Health Aide control declined 22.3%, rather than remaining roughly flat as originally expected. This means broader labor-market or occupation-specific forces cannot be ruled out.

A same-source CPS check on Software Developers, the occupation most directly cited as the primary study's clearest case of early-career disruption (Brynjolfsson et al., 2026), found only a 1.6% decline in this pull, compared with roughly 18% in the primary study's published series over a comparable window. The estimates use different sources and methodologies—CPS survey data versus ADP payroll data—which may contribute to the

difference, but this paper does not attempt to reconcile them. The discrepancy is reported rather than smoothed over.

The second test asks a different question: what happens if reduced junior hiring persists, regardless of exactly how much of the initial decline is attributable to AI? A cohort model tracks junior hires into the future mid-level workforce. The primary case assumes three years for a junior worker to progress to mid-level, with five years as a sensitivity case. Under the Depressed scenario, junior hiring remains at its current reduced level through 2030; under Recovery, it returns to its pre-2022 trend by 2028.

The timing result supports H2. Under Depressed, the modeled mid-level shortage begins in 2026 with a three-year transition and 2028 with a five-year transition. Under Recovery, the gap narrows but remains open through 2035.

These figures are scenario outputs, not forecasts: even if firms eventually restore junior hiring, new hires cannot immediately become the experienced workers earlier cohorts would have produced.

H1 does not establish that Graphic Designers experienced a larger AI-driven decline, and the control weakens a clean causal interpretation. H2 does not establish that Graphic Designers are experiencing the depressed scenario; it takes the broader early-career employment effect identified in the primary study (Brynjolfsson et al., 2026) as an input and asks what follows if that effect persists. H2 tests pipeline timing, not the occupation-specific mechanism tested by H1.

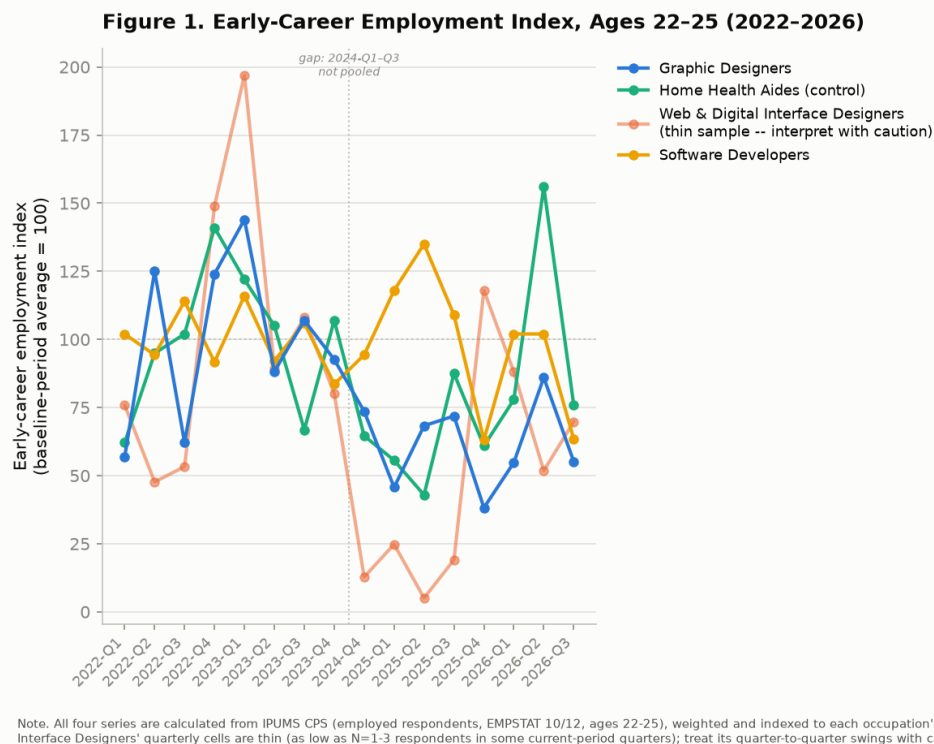
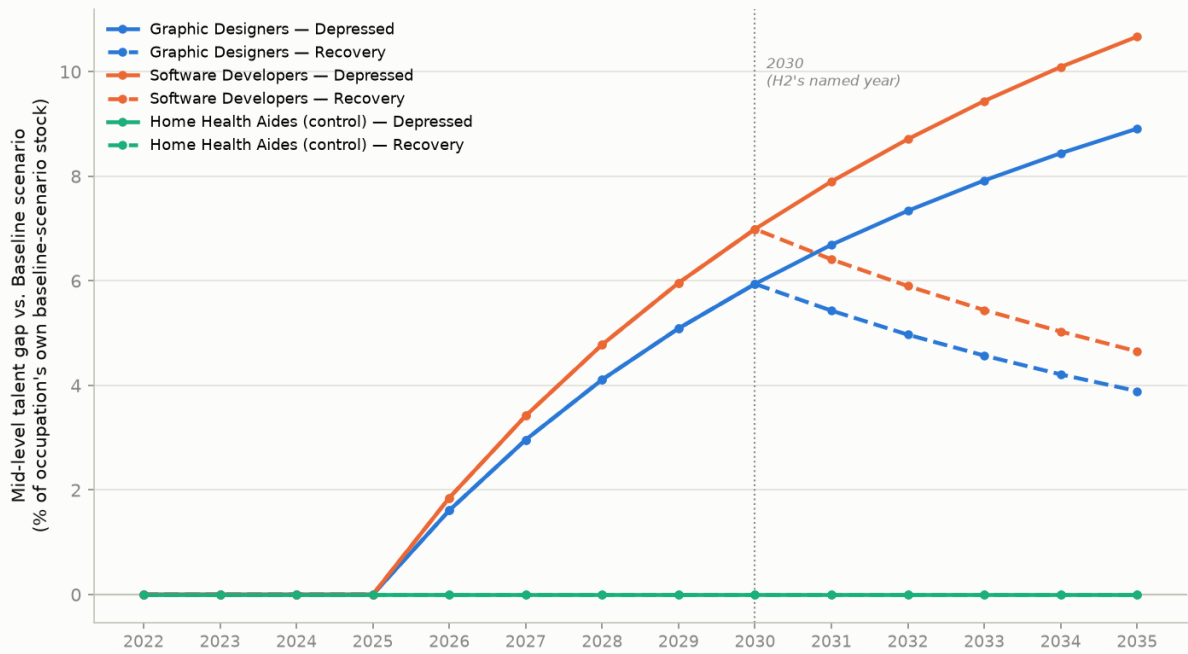


Figure 1. Early-career employment index, 2022–2026. Figure 1 compares early-career employment trajectories across the four occupations. It is descriptive rather than causal: it shows whether occupations experienced similar or different changes among workers ages 22–25 during the study period. The broader primary-study evidence addresses the distinction between early-career and experienced-worker outcomes. *Note.* All four series are calculated from IPUMS CPS data using the same methodology, weighted and indexed to each occupation's baseline-period average. Interface Designer quarterly samples are thin, so its swings should be interpreted cautiously.

Figure 2. Projected Mid-Level Talent Gap, Three Occupations (N=3, 2022-2035)



Note. Output of the cohort model (training lag N=3, the primary case) run separately for each occupation's own real inputs -- see capabilities/economic-research/pipeline-model-spec.md. Home Health Aides' Depressed_Hiring_Pct (1.0, no change) is inferred from the primary paper treating the least-exposed quintile as the omitted regression reference category, not read directly off a table cell -- flagged as an assumption, not a confirmed figure. Scenario outputs under explicit assumptions, not a forecast.

Figure 2 (appendix candidate). Projected mid-level talent gap, three occupations, 2022–2035. Output of the cohort model (training lag N=3, the primary case) run separately for each occupation's own real inputs. Gap is expressed as a percentage of each occupation's own baseline-scenario stock. Home Health Aides' Depressed_Hiring_Pct (1.0, no change) is inferred from the primary paper treating the least-exposed quintile as the omitted regression reference category, not read directly off a table cell — flagged as an assumption, not a confirmed figure.

Policy Recommendation

The case for intervention does not rest on H1 proving that AI caused the Graphic Designer decline. It rests on the broader evidence that AI-exposed occupations have experienced disproportionate early-career employment declines (Brynjolfsson et al., 2026) and on the economic logic that firms may underinvest in training when they cannot capture its full return. The recommendation is therefore conditional: where AI-driven reductions in junior work create a training shortfall, collective investment can address the market failure more directly than restricting AI adoption.

First, establish a mandatory training levy on firms within the affected sector. Contributions would fund shared early-career training and prevent firms from free-riding on other employers' investments.

Second, policymakers should provide the legal foundation and oversight, while firms and the sector body determine training priorities and delivery.

Third, complement the levy with targeted apprenticeship funding or entry-level hiring credits to reduce the immediate cost and risk of hiring juniors. Firms should also redesign junior roles around AI oversight, judgment, and development, giving early-career workers responsibility for directing and correcting AI-generated work.

The strongest objection is that the market may eventually self-correct—or that the observed decline may not be sufficiently AI-specific to justify intervention. The evidence here cannot rule out the latter. But where AI has genuinely removed training opportunities, waiting for a mid-level shortage to trigger new hiring creates a timing problem: newly hired juniors still require years to become experienced workers.

The goal, therefore, is not to stop the market from adopting AI. It is to preserve the training function of the first rung while allowing the work on that rung to change.

References

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